

Difficulty in Accessing Da... stores_file x

Limit to 1000 r

```
1 • SELECT * FROM bad_retaildb.stores_file;
```

Result Grid Filter Rows: Edit: Export/In

store_id	store_name	region	city
1	Yangon Central Store	South	Yangon
2	Mandalay Plaza Store	Central	Mandalay
3	Naypyitaw Junction Store	Central	Naypyitaw
4	Mawlamyine Market Store	South	Mawlamyine
5	Taunggyi Hill Store	East	Taunggyi
6	Myitkyina Star Store	North	Myitkyina
NULL	NULL	NULL	NULL

Difficulty in Accessing Da... products_file x

Limit to 1000 r

```
1 • SELECT * FROM bad_retaildb.products_file;
```

Result Grid Filter Rows: Edit: Export/Import

product_id	product_name	category	unit_price
101	Rice Bag	Grocery	45000.00
102	Cooking Oil	Grocery	12000.00
103	Instant Noodles	Grocery	800.00
104	Soft Drink	Beverage	1500.00
105	Shampoo	Personal Care	7000.00
106	Soap	Personal Care	2500.00
NULL	NULL	NULL	NULL

Difficulty in Accessing Da...						
sales_file x						
Limit to 1000 r						
1 • SELECT * FROM bad_retaildb.sales_file;						
Result Grid						
Filter Rows:						
Edit:						
Export/Import:						
Wrap Ce						
	sale_id	sale_date	store_id	product_id	quantity	total_amount
▶	1	2025-10-05	1	101	10	450000.00
	2	2025-10-12	2	102	20	240000.00
	3	2025-10-18	3	103	100	80000.00
	4	2025-10-22	4	104	60	90000.00
	5	2025-10-25	5	105	15	105000.00
	6	2025-10-28	6	106	30	75000.00
	7	2025-11-03	1	102	25	300000.00
	8	2025-11-07	2	101	8	360000.00
	9	2025-11-11	3	104	70	105000.00
	10	2025-11-15	4	103	120	96000.00
	11	2025-11-19	5	106	40	100000.00
	12	2025-11-24	6	105	12	84000.00
	13	2025-12-02	1	103	150	120000.00
	14	2025-12-06	2	104	80	120000.00
	15	2025-12-10	3	101	9	405000.00
	16	2025-12-14	4	102	18	216000.00
	17	2025-12-18	5	105	20	140000.00
	18	2025-12-22	6	106	35	87500.00
	19	2026-01-05	1	104	90	135000.00
	20	2026-01-08	2	103	140	112000.00
	21	2026-01-12	3	102	22	264000.00
	22	2026-01-16	4	101	7	315000.00
	23	2026-01-20	5	106	55	137500.00
	24	2026-01-27	6	105	10	70000.00
	25	2026-02-03	1	105	18	126000.00
	26	2026-02-07	2	106	50	125000.00
	27	2026-02-11	3	104	75	112500.00
	28	2026-02-15	4	103	160	128000.00
	29	2026-02-19	5	101	11	495000.00
	30	2026-02-24	6	106	16	102000.00

sales_file1 x

Difficulty in Accessing Da...

Limit to 1000 r

```

130  -- Without a proper integrated design, report logic becomes custom and repetitive
131  •  SELECT
132      s.sale_date,
133      s.store_id,
134      st.store_name,
135      st.region,
136      s.total_amount
137  FROM sales_file s
138  JOIN stores_file st ON s.store_id = st.store_id
139  WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
140  ORDER BY st.region, s.sale_date;

```

Result Grid
Filter Rows:
Export:
Wrap Cell Content:

	sale_date	store_id	store_name	region	total_amount
▶	2025-10-12	2	Mandalay Plaza Store	Central	240000.00
	2025-10-18	3	Naypyitaw Junction Store	Central	80000.00
	2025-11-07	2	Mandalay Plaza Store	Central	360000.00
	2025-11-11	3	Naypyitaw Junction Store	Central	105000.00
	2025-12-06	2	Mandalay Plaza Store	Central	120000.00
	2025-12-10	3	Naypyitaw Junction Store	Central	405000.00
	2026-01-08	2	Mandalay Plaza Store	Central	112000.00
	2026-01-12	3	Naypyitaw Junction Store	Central	264000.00
	2026-02-07	2	Mandalay Plaza Store	Central	125000.00
	2026-02-11	3	Naypyitaw Junction Store	Central	112500.00
	2026-03-08	2	Mandalay Plaza Store	Central	98000.00
	2026-03-12	3	Naypyitaw Junction Store	Central	144000.00
	2025-10-25	5	Taunggyi Hill Store	East	105000.00
	2025-11-19	5	Taunggyi Hill Store	East	100000.00
	2025-12-18	5	Taunggyi Hill Store	East	140000.00
	2026-01-20	5	Taunggyi Hill Store	East	137500.00
	2026-02-19	5	Taunggyi Hill Store	East	495000.00
	2026-03-20	5	Taunggyi Hill Store	East	252000.00
	2025-10-28	6	Myitkyina Star Store	North	75000.00
	2025-11-24	6	Myitkyina Star Store	North	84000.00

Result 1

Difficulty in Accessing Da... stores x

Limit to 1000 r

1 • **SELECT * FROM** good_retaildb.stores;

Result Grid Filter Rows: Edit: Export/In

	store_id	store_name	region	city
▶	1	Yangon Central Store	South	Yangon
	2	Mandalay Plaza Store	Central	Mandalay
	3	Naypyitaw Junction Store	Central	Naypyitaw
	4	Mawlamyine Market Store	South	Mawlamyine
	5	Taunggyi Hill Store	East	Taunggyi
	6	Myitkyina Star Store	North	Myitkyina
*	NULL	NULL	NULL	NULL

Difficulty in Accessing Da... products x

Limit to 1000 r

1 • **SELECT * FROM** good_retaildb.products;

Result Grid Filter Rows: Edit: Export/In

	product_id	product_name	category	unit_price
▶	101	Rice Bag	Grocery	45000.00
	102	Cooking Oil	Grocery	12000.00
	103	Instant Noodles	Grocery	800.00
	104	Soft Drink	Beverage	1500.00
	105	Shampoo	Personal Care	7000.00
	106	Soap	Personal Care	2500.00
*	NULL	NULL	NULL	NULL

Difficulty in Accessing Da... sales x

Limit to 1000 r

1 • **SELECT * FROM** good_retaildb.sales;

Result Grid						
Filter Rows:						
Edit:						
Export/Import:						
	sale_id	sale_date	store_id	product_id	quantity	total_amount
▶	1	2025-10-05	1	101	10	450000.00
	2	2025-10-12	2	102	20	240000.00
	3	2025-10-18	3	103	100	80000.00
	4	2025-10-22	4	104	60	90000.00
	5	2025-10-25	5	105	15	105000.00
	6	2025-10-28	6	106	30	75000.00
	7	2025-11-03	1	102	25	300000.00
	8	2025-11-07	2	101	8	360000.00
	9	2025-11-11	3	104	70	105000.00
	10	2025-11-15	4	103	120	96000.00
	11	2025-11-19	5	106	40	100000.00
	12	2025-11-24	6	105	12	84000.00
	13	2025-12-02	1	103	150	120000.00
	14	2025-12-06	2	104	80	120000.00
	15	2025-12-10	3	101	9	405000.00
	16	2025-12-14	4	102	18	216000.00
	17	2025-12-18	5	105	20	140000.00
	18	2025-12-22	6	106	35	87500.00
	19	2026-01-05	1	104	90	135000.00
	20	2026-01-08	2	103	140	112000.00
	21	2026-01-12	3	102	22	264000.00
	22	2026-01-16	4	101	7	315000.00
	23	2026-01-20	5	106	55	137500.00
	24	2026-01-27	6	105	10	70000.00
	25	2026-02-03	1	105	18	126000.00
	26	2026-02-07	2	106	50	125000.00
	27	2026-02-11	3	104	75	112500.00
	28	2026-02-15	4	103	160	128000.00
	29	2026-02-19	5	101	11	495000.00
	30	2026-02-24	6	102	16	192000.00
	31	2026-03-04	1	106	60	150000.00

sales 1 x

```
187
188 -- Workflow for DBMS Solution
189 -- the manager's question can be answered directly by SQL.
190 -- Sales report for last 6 months by region
191 • SELECT
192     st.region,
193     SUM(s.total_amount) AS total_sales
194 FROM sales s
195 JOIN stores st ON s.store_id = st.store_id
196 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
197 GROUP BY st.region
198 ORDER BY total_sales DESC;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	region	total_sales
▶	South	2268500.00
	Central	2165500.00
	East	1229500.00
	North	1093500.00

Difficulty in Accessing Da... x

Limit to 1000 r

```

200  -- Monthly sales report by region
201  •  SELECT
202      DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month,
203      st.region,
204      SUM(s.total_amount) AS total_sales
205  FROM sales s
206  JOIN stores st ON s.store_id = st.store_id
207  WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
208  GROUP BY DATE_FORMAT(s.sale_date, '%Y-%m'), st.region
209  ORDER BY sales_month, st.region;

```

Result Grid
Filter Rows:
Export:
Wrap Cell Content:

	sales_month	region	total_sales
▶	2025-10	Central	320000.00
	2025-10	East	105000.00
	2025-10	North	75000.00
	2025-10	South	540000.00
	2025-11	Central	465000.00
	2025-11	East	100000.00
	2025-11	North	84000.00
	2025-11	South	396000.00
	2025-12	Central	525000.00
	2025-12	East	140000.00
	2025-12	North	87500.00
	2025-12	South	336000.00
	2026-01	Central	376000.00
	2026-01	East	137500.00
	2026-01	North	70000.00
	2026-01	South	450000.00
	2026-02	Central	237500.00
	2026-02	East	495000.00
	2026-02	North	192000.00
	2026-02	South	251000.00

Result 3 x

Difficulty in Accessing Da... x

211 -- Ad-hoc report: sales by region and product category
212 • **SELECT**
213 st.region,
214 p.category,
215 SUM(s.total_amount) **AS** total_sales
216 **FROM** sales s
217 **JOIN** stores st **ON** s.store_id = st.store_id
218 **JOIN** products p **ON** s.product_id = p.product_id
219 **WHERE** s.sale_date **BETWEEN** '2025-10-01' **AND** '2026-03-31'
220 **GROUP BY** st.region, p.category
221 **ORDER BY** st.region, total_sales **DESC**;

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	region	category	total_sales
▶	Central	Grocery	1605000.00
	Central	Beverage	337500.00
	Central	Personal Care	223000.00
	East	Grocery	747000.00
	East	Personal Care	482500.00
	North	Grocery	777000.00
	North	Personal Care	316500.00
	South	Grocery	1625000.00
	South	Beverage	367500.00
	South	Personal Care	276000.00

242

243 • **SELECT * FROM vw_sales_reporting;**

244

245 -- Test Case Scripts

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	sale_id	sale_date	store_id	store_name	region	city	product_id	product_name	category	quantity	total_amount
▶	1	2025-10-05	1	Yanqon Central Store	South	Yanqon	101	Rice Bag	Grocery	10	450000.00
	7	2025-11-03	1	Yanqon Central Store	South	Yanqon	102	Cooking Oil	Grocery	25	300000.00
	13	2025-12-02	1	Yanqon Central Store	South	Yanqon	103	Instant Noodles	Grocery	150	120000.00
	19	2026-01-05	1	Yanqon Central Store	South	Yanqon	104	Soft Drink	Beverage	90	135000.00
	25	2026-02-03	1	Yanqon Central Store	South	Yanqon	105	Shampoo	Personal Care	18	126000.00
	31	2026-03-04	1	Yanqon Central Store	South	Yanqon	106	Soap	Personal Care	60	150000.00
	2	2025-10-12	2	Mandalay Plaza Store	Central	Mandalay	102	Cooking Oil	Grocery	20	240000.00
	8	2025-11-07	2	Mandalay Plaza Store	Central	Mandalay	101	Rice Bag	Grocery	8	360000.00
	14	2025-12-06	2	Mandalay Plaza Store	Central	Mandalay	104	Soft Drink	Beverage	80	120000.00
	20	2026-01-08	2	Mandalay Plaza Store	Central	Mandalay	103	Instant Noodles	Grocery	140	112000.00
	26	2026-02-07	2	Mandalay Plaza Store	Central	Mandalay	106	Soap	Personal Care	50	125000.00
	32	2026-03-08	2	Mandalay Plaza Store	Central	Mandalay	105	Shampoo	Personal Care	14	98000.00
	3	2025-10-18	3	Naypyitaw Junction ...	Central	Naypyitaw	103	Instant Noodles	Grocery	100	80000.00
	9	2025-11-11	3	Naypyitaw Junction ...	Central	Naypyitaw	104	Soft Drink	Beverage	70	105000.00
	15	2025-12-10	3	Naypyitaw Junction ...	Central	Naypyitaw	101	Rice Bag	Grocery	9	405000.00
	21	2026-01-12	3	Naypyitaw Junction ...	Central	Naypyitaw	102	Cooking Oil	Grocery	22	264000.00
	27	2026-02-11	3	Naypyitaw Junction ...	Central	Naypyitaw	104	Soft Drink	Beverage	75	112500.00
	33	2026-03-12	3	Naypyitaw Junction ...	Central	Naypyitaw	103	Instant Noodles	Grocery	180	144000.00
	4	2025-10-22	4	Mawlamyine Market ...	South	Mawlam...	104	Soft Drink	Beverage	60	90000.00
	10	2025-11-15	4	Mawlamyine Market ...	South	Mawlam...	103	Instant Noodles	Grocery	120	96000.00
	16	2025-12-14	4	Mawlamyine Market ...	South	Mawlam...	102	Cooking Oil	Grocery	18	216000.00
	22	2026-01-16	4	Mawlamyine Market ...	South	Mawlam...	101	Rice Bag	Grocery	7	315000.00
	28	2026-02-15	4	Mawlamyine Market ...	South	Mawlam...	103	Instant Noodles	Grocery	160	128000.00
	34	2026-03-16	4	Mawlamyine Market ...	South	Mawlam...	104	Soft Drink	Beverage	95	142500.00
	5	2025-10-25	5	Taunggyi Hill Store	East	Taunggyi	105	Shampoo	Personal Care	15	105000.00
	11	2025-11-19	5	Taunggyi Hill Store	East	Taunggyi	106	Soap	Personal Care	40	100000.00
	17	2025-12-18	5	Taunggyi Hill Store	East	Taunggyi	105	Shampoo	Personal Care	20	140000.00
	23	2026-01-20	5	Taunggyi Hill Store	East	Taunggyi	106	Soap	Personal Care	55	137500.00
	29	2026-02-19	5	Taunggyi Hill Store	East	Taunggyi	101	Rice Bag	Grocery	11	495000.00

vw_sales_reporting5 x

Difficulty in Accessing Da... x

```
248 • USE bad_retaildb;
249 • SELECT
250     sale_id,
251     sale_date,
252     store_id,
253     total_amount
254 FROM sales_file
255 WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31';
```

Result Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Cont

	sale_id	sale_date	store_id	total_amount
▶	1	2025-10-05	1	450000.00
	2	2025-10-12	2	240000.00
	3	2025-10-18	3	80000.00
	4	2025-10-22	4	90000.00
	5	2025-10-25	5	105000.00
	6	2025-10-28	6	75000.00
	7	2025-11-03	1	300000.00
	8	2025-11-07	2	360000.00
	9	2025-11-11	3	105000.00
	10	2025-11-15	4	96000.00
	11	2025-11-19	5	100000.00
	12	2025-11-24	6	84000.00
	13	2025-12-02	1	120000.00
	14	2025-12-06	2	120000.00
	15	2025-12-10	3	405000.00
	16	2025-12-14	4	216000.00
	17	2025-12-18	5	140000.00
	18	2025-12-22	6	87500.00
	19	2026-01-05	1	135000.00
	20	2026-01-08	2	112000.00
	21	2026-01-12	3	264000.00
	22	2026-01-16	4	315000.00
	23	2026-01-20	5	137500.00
	24	2026-01-23	6	70000.00

sales_file 6 x

Difficulty in Accessing Da...

Limit to 1000 r

```

259 • SELECT
260     s.sale_id,
261     s.sale_date,
262     st.region,
263     s.total_amount
264 FROM sales_file s
265 JOIN stores_file st ON s.store_id = st.store_id
266 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31';

```

Result Grid
Filter Rows:
Export:
Wrap Cell Content:

	sale_id	sale_date	region	total_amount
▶	1	2025-10-05	South	450000.00
	2	2025-10-12	Central	240000.00
	3	2025-10-18	Central	80000.00
	4	2025-10-22	South	90000.00
	5	2025-10-25	East	105000.00
	6	2025-10-28	North	75000.00
	7	2025-11-03	South	300000.00
	8	2025-11-07	Central	360000.00
	9	2025-11-11	Central	105000.00
	10	2025-11-15	South	96000.00
	11	2025-11-19	East	100000.00
	12	2025-11-24	North	84000.00
	13	2025-12-02	South	120000.00
	14	2025-12-06	Central	120000.00
	15	2025-12-10	Central	405000.00
	16	2025-12-14	South	216000.00
	17	2025-12-18	East	140000.00
	18	2025-12-22	North	87500.00
	19	2026-01-05	South	135000.00
	20	2026-01-08	Central	112000.00
	21	2026-01-12	Central	264000.00
	22	2026-01-16	South	315000.00
	23	2026-01-20	East	137500.00
	24	2026-01-27	North	70000.00

Result 7

Difficulty in Accessing Da... x

Limit to 1000 r

```

268 -- Test Case 3: DBMS quickly answers manager's question
269 • USE good_retaildb;
270 • SELECT
271     st.region,
272     SUM(s.total_amount) AS total_sales
273 FROM sales s
274 JOIN stores st ON s.store_id = st.store_id
275 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
276 GROUP BY st.region
277 ORDER BY total_sales DESC;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	region	total_sales
▶	South	2268500.00
	Central	2165500.00
	East	1229500.00
	North	1093500.00

Difficulty in Accessing Da... x

Limit to 1000 r

```

278
279 -- Test Case 4: Ad-hoc reporting
280
281 • SELECT
282     region,
283     COUNT(*) AS transaction_count,
284     SUM(total_amount) AS total_sales
285 FROM vw_sales_reporting
286 WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31'
287 GROUP BY region
288 ORDER BY total_sales DESC;
289

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	region	transaction_count	total_sales
▶	South	12	2268500.00
	Central	12	2165500.00
	East	6	1229500.00
	North	6	1093500.00

Difficulty in Accessing Da... x



Limit to 1000 r



289

290 -- Test Case 5: Monthly trend by region

291 • **SELECT**

292 DATE_FORMAT(sale_date, '%Y-%m') AS sales_month,

293 region,

294 SUM(total_amount) AS total_sales

295 **FROM** vw_sales_reporting

296 **WHERE** sale_date **BETWEEN** '2025-10-01' **AND** '2026-03-31'

297 **GROUP BY** DATE_FORMAT(sale_date, '%Y-%m'), region

298 **ORDER BY** sales_month, region;

299

Result Grid



Filter Rows:

Export: 

Wrap Cell Content: 

	sales_month	region	total_sales
▶	2025-10	Central	320000.00
	2025-10	East	105000.00
	2025-10	North	75000.00
	2025-10	South	540000.00
	2025-11	Central	465000.00
	2025-11	East	100000.00
	2025-11	North	84000.00
	2025-11	South	396000.00
	2025-12	Central	525000.00
	2025-12	East	140000.00
	2025-12	North	87500.00
	2025-12	South	336000.00
	2026-01	Central	376000.00
	2026-01	East	137500.00
	2026-01	North	70000.00
	2026-01	South	450000.00
	2026-02	Central	237500.00
	2026-02	East	495000.00
	2026-02	North	192000.00
	2026-02	South	254000.00
	2026-03	Central	242000.00

Result 11 x